Appendix

Table 1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1-P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4-P5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6-P7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	P7, appendix 2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P8-P9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the	P8-P9

Section and Topic	Item #	Checklist item	Location where item is reported
		process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P8-P9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P8-P9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P8-P9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P9-P10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P9-P10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P9-P10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P9-P10
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P9-P10
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P10-P11
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	P11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P19 appendix 4
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P10-P18, appendix 3
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P10-P20
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P19-P20
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	P19
Certainty of	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P15-P20

Section and Topic	Item #	Checklist item	Location where item is reported
evidence			
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P21
	23b	Discuss any limitations of the evidence included in the review.	P24
	23c	Discuss any limitations of the review processes used.	N/A
	23d	Discuss implications of the results for practice, policy, and future research.	P22-P25
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P6-P7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	P6-P7
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P25
Competing interests	26	Declare any competing interests of review authors.	P25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P25

Table 2 Search strategies to identify studies reporting the prevalence status, treatment and outcomes of tuberculosis and COVID-19

Database searched: PubMed

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	("covid-19"[MeSH Terms] OR "sars-cov-2"[MeSH Terms]) AND "tuberculosis"[MeSH Terms]	244
2	((covid-19[Title/Abstract]) OR (Sars-cov-2[Title/Abstract])) AND((TB[Title/Abstract]) OR (tuberculos*[Title/Abstract]))	523
3	#1 OR #2	565

Database searched: Web of Science

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	((TS=(covid-19) OR TS=(sars-cov-2) OR AB=(covid-19) OR AB=(sars-cov-2)) OR (TS=(covid-19)) OR (TS=(sars-cov-2)) OR (AB=(covid-19))) AND (TS=(tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (TS=(Mycobacterium tuberculosis)) OR (AB=(tuberculo*)) OR (AB=(TB)) OR (AB=(Mycobacterium tuberculosis)) OR (AB=(M.tuberculosis)) OR (AB=(tubercle bacillus)))	644

Database searched: ProQuest

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	su(COVID-19 OR SARS-COV-2) AND su(TB OR tuberculo* OR (Mycobacterium tuberculosis) OR (M.tuberculosis))	682

Database searched: Scopus

Date: Mars 19th, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	(TITLE-ABS-KEY (covid-19 OR sars-cov-2) AND TITLE-ABS-KEY (tb OR tuberculos* OR (mycobacterium AND tuberculosis) OR (m.tuberculosis)))	839

Database searched: Cochrane database

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	MeSH descriptor: [COVID-19] explode all trees	257
2	MeSH descriptor: [SARS-COV-2] explode all trees	204
3	MeSH descriptor: [tuberculosis] explode all trees	2,029
4	(#1 OR #2) AND 3	1

Database searched: Embase

Date: Mars 19, 2021

Limits: Nov. 1st, 2019 to Mar. 19th, 2021

#	Searches	Results
1	('covid 19':ab,ti OR 'sars cov 2':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	81,923
2	(tuberculosis:ab,ti OR 'mycobacterium tuberculosis':ab,ti OR tuberculo*:ab,ti OR 'tb':ab,ti OR 'm.tuberculosis':ab,ti OR 'tubercle bacillus':ab,ti) AND [01-11-2019]/sd NOT [20-3-2021]/sd	19,510
3	#1 AND #2	367

Table 3 Characteristics of reported TB and COVID-19 coinfection patients (n=55)

Basic information	Number (%)
Gender	
female	12 (21.8)
male	43 (78.2)
Age	
< 20	3 (5.5)
20-39	23 (41.8)
40-59	22 (40.0)
≥60	7 (12.7)
Comorbidity	
diabetes mellitus	8 (14.5)
hypertension	5 (9.1)
HIV	4 (7.3)
Type of TB	
active TB	50 (90.9)
post TB	4 (7.3)
latent TB	1 (1.8)
TB site	
pulmonary TB	47 (85.5)
central nervous system involvement	2 (3.6)
tuberculous meningitis	1 (1.8)
tuberculous pleuritis	1 (1.8)
peritoneal TB	1 (1.8)
not reported	6 (10.9)

BCG Vaccination	
yes	2 (3.6)
not reported	53 (96.4)
Drug resistance	
no	15 (27.3)
yes	2 (3.6)
likely	1 (1.8)
not reported	37 (67.3)
Infection order	
TB before COVID-19	17
COVID-19 before TB	0
not reported	38
Symptomatic COVID-19	
yes	55 (100.0)
no	0 (0.0)

Table 4 Detailed basic information of included case reports (n=35)

Author (Year)	Country	Cases number	Age	Gender	Comorbidity	BCG Vaccination	TB site	Drug resistance	Active TB	TB before COVID-19	Symptomatic COVID-19

Sarma 2020[1]	India	1	53	female	chronic kidney disease (CKD) stage 4, type 2 diabetes mellitus	/	pulmonary TB	likely	yes	yes	yes
Singh 2020[2]	India	1	76	female	/	/	pulmonary TB	/	yes	/	yes
Kumar 2020[3]	India	1	38	male	/	/	pulmonary TB	/	yes	/	yes
Ata 2020[4]	India	1	28	male	no HIV	/	pulmonary TB central nervous system involvement	/	yes	/	yes
Yadav 2020[5]	India	1	43	male	non-diabetes	/	pulmonary TB	no	yes	/	yes
Baskara 2021[6]	Indonesian	1	42	male	diabetes mellitus	/	pulmonary TB	no	yes	yes	yes
Wang 2020[7]	China	1	45	male	/	/	tuberculous meningitis	/	yes	/	yes
			67	male	hypertension	/	/	/	no	yes	yes
			26	male	/	/	/	/	no	yes	yes
He 2020[8]	China	3	76	male	exfoliative dermatitis, Arthrolithiasis	/	/	/	no	yes	yes
Cao 2020[9]	China	1	47	female	/	/	pulmonary TB	/	yes	yes	yes

			50	male	anemia, hypoproteinemia	/	pulmonary TB	/	yes	yes	yes
Yao 2020[10]	China	3	44	male	diabetes, pleural effusion	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
			57	male	diabetes	/	pulmonary TB	/	yes	yes	yes
			48	male	/	/	/	/	no	/	yes
			26	male	/	/	pulmonary TB	yes	yes	yes	yes
Liu 2020[11]	China	3	46	male	/	/	pulmonary TB, tuberculous pleuritis	/	no	yes	yes
Khayat 2021[12]	Saudi Arabia	1	40	female	no HIV and hepatitis	/	pulmonary TB	no	yes	/	yes
Faqihi 2020[13]	Saudi Arabia	1	60	male	hypertensive and diabetic	/	pulmonary TB	no	yes	/	yes
			34	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
Yousaf 2020[14]	Qatar	6	32	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			50	male	no HIV, hepatitis B, and hepatitis C	/	pulmonary TB	/	yes	/	yes

					infection; have diabetes						
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			27	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	/	yes	/	yes
			35	male	no HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB, tuberculous pleuritis	/	yes	/	yes
AlKhateeb 2020[15]	Qatar	1	28	male	/	/	pulmonary TB	/	yes	/	yes
Tham 2020[16]	Singapore	4	32	male	/	/	pulmonary TB	/	yes	/	yes
			33	male	/	/	pulmonary TB	/	yes	/	yes
			22	male	/	/	pulmonary TB	/	yes	/	yes
			40	male	/	/	pulmonary TB	/	yes	/	yes
Cinar 2020[17]	Turkey	1	55	male	/	/	/	/	yes	/	yes
Sarinoglu 2020[18]	Turkey	2	77	female	diabetes mellitus, hypertension, chronic obstructive lung disease, chronic kidney failure	/	pulmonary TB	/	yes	/	yes
Lopinto	France	1	39	female	/	/	pulmonary TB	/	yes	/	yes
			58	male	pulmonary	/	pulmonary TB	/	yes	yes	yes

2020[19]					embolism						
Musso 2021[20]	Italy	1	45	male	No HIV	/	pulmonary TB	no	yes	yes	yes
Luciani 2020[21]	Italy	1	32	female	No HIV	yes	pulmonary TB	/	yes	/	yes
Bouaré 2020[22]	Morocco	1	32	female	/	/	pulmonary TB	/	yes	/	yes
Essajee 2020[23]	South Africa	1	2	female	/	yes	pulmonary TB	no	yes	/	yes
Pillay 2021[24]	South Africa	1	44	female	HIV, pneumocystis iirovecii pneumonia, community-acquired pneumonia, pulmonary embolism	/	pulmonary TB	/	yes	/	yes
Agada 2021[25]	Nigeria	1	50	male	No HIV, hepatitis B, and hepatitis C infection	/	pulmonary TB	no	yes	/	yes
Gbenga 2020[26]	Nigeria	2	33	male	/	/	pulmonary TB	/	yes	/	yes
			30	male	/	/	pulmonary TB	/	yes	/	yes
Rivas 2020[27]	Panama	2	29	male	pleural empyema	/	pulmonary TB	no	yes	/	yes
			53	male	HIV	/	pulmonary TB	/	yes	yes	yes
Farias 2020[28]	Brazil	2	39	male	HIV, hepatitis B	/	pulmonary TB	no	yes	yes	yes
			43	male	HIV, anxiety	/	pulmonary TB	no	yes	/	yes

Pinheiro 2020[29]	Brazil	1	68	male	disorder diabetes, hypertension, chronic liver disease secondary to schistosomiasis	/	pulmonary TB	no	yes	/	yes
Gerstein 2021[30]	USA	1	49	male	cirrhosis, portal hypertension	/	pulmonary TB, peritoneal TB	/	yes	/	yes
Cutler 2020[31]	USA	1	61	male	Parkinson's disease	/	pulmonary TB	/	yes	/	yes
Sasson 2020[32]	USA	1	44	male	no HIV; hypertension, type II diabetes mellitus, a previous stroke secondary to a left atrial thrombus	/	/	no	yes	/	yes
Freij 2020[33]	USA	1	5	female	no HIV	/	central nervous system involvement	/	yes	/	yes
Vanzetti 2020[34]	Argentina	3	33	female	/	/	pulmonary TB	no	yes	/	yes
			34	male	/	/	pulmonary TB	no	yes	yes	yes
			17	male	/	/	pulmonary TB	no	yes	/	yes
Vilbrun 2020[35]	Haiti	1	26	male	no HIV	/	pulmonary TB	yes	yes	yes	yes

/: No information provided

Table 5 Detailed treatment information of included cases (n=55)

Author (Year)	Cases number	ICU	Treatment before hospitalization	Treatment in hospital	Treatment after discharge	Post-intervention clinical condition	Outcome	Days to die or discharge
Sarma 2020[1]	1	yes	/	**Supportive treatment:** oxygen support (intubated) **Antibiotic:** ampicillin, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** vitamin C, midazolam, fentanyl	/	/	resolved	/
Singh 2020[2]	1	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** N-acetyl cysteine, methylprednisolone, vitamin C	/	/	on treatment	/

Kumar 2020[3]	1	no	/	**Supportive treatment:** oxygen support, mechanical ventilation **Antibiotic:** ATT, other antibiotic (no detail) **Antiviral drug:** hydroxychloroquine sulphate **Other treatment:** /	/	/	died	/
Ata 2020[4]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** steroids	rifampicin, ethambutol, and pyrazinamide	/	resolved	28

Yadav 2020[5]	1	no	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, ethambutol, and pyrazinamide) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
Baskara 2021[6]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin, ceftriaxone, ceftazidime **Antiviral drug:** hydroxychloroquine, oseltamivir **Other treatment:** anti-coagulant drugs	treatment for diabetes mellitus	/	resolved	13

Wang 2020[7]	1	no	/	**Supportive treatment:** / **Antibiotic:** rifampicin, amikacin, moxifloxacin **Antiviral drug:** arbidol **Other treatment:** tolvaptan, aspirin, mannitol, potassium supplement, sodium supplement	/	/	died	14
He 2020[8]	3	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	37

no	/	**Supportive treatment:** / **Antibiotic:** antibiotic (no details and not ATT) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** probiotics, traditional Chinese medicine	/	skin rashes with pruritus	ninth day after discharge, returned to the hospital because of positive recurrence of SARS-CoV-2 RNA	28
no	/	**Supportive treatment:** nasal cannula oxygen support **Antibiotic:** ATT, antibiotic (no details) **Antiviral drug:** lopinavir, ritonavir, arbidol **Other treatment:** methylprednisolone, glucocorticoid, probiotics, traditional Chinese medicine, immunoglobulin	/	/	resolved	24

Cao 2020[9]	1	no	/	**Supportive treatment:** noninvasive ventilation support **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), moxifloxacin hydrochloride **Antiviral drug:** lopinavir and ritonavir, umifenovir **Other treatment:** polyene phosphati-dylcholine, reduced glutathione	/	Hepatic injury during the treatment	resolved	More than 14 days
Yao 2020[10]	3	no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** ATT **Antiviral drug:** / **Other treatment:** corticosteroid	/	/	died	22

		no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** acarbose, metformin	ATT	/	resolved	26
		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** antibiotics (no details) **Antiviral drug:** antiviral drugs (no details) **Other treatment:** corticosteroids, immunoglobulin	/	/	resolved	27
Liu 2020[11]	3	no	/	**Supportive treatment:** extracorporeal membrane oxygenation, haemoperfusion **Antibiotic:** moxifloxacin, linezolid, no ATT **Antiviral drug:** arbidol **Other treatment:**	/	/	resolved	14

		caspofungin, thymopentin				
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin, linezolid, cycloserine, clofazimine, pyrazinamide **Antiviral drug:** arbidol **Other treatment:** /	ATT	/	resolved	14
no	/	**Supportive treatment:** nasal high-flow oxygen therapy **Antibiotic:** moxifloxacin **Antiviral drug:** arbidol **Other treatment:** /	/	/	resolved	9

Khayat 2021[12]	1	no	/	/	/	/	/	/
Faqihi 2020[13]	1	yes	/	**Supportive treatment:** respiratory support **Antibiotic:** moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir, ribavirin **Other treatment:** dexamethasone, prophylactic anticoagulation	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), isoniazid and rifampicin (later)	/	resolved	27
Yousaf 2020[14]	6	no	/	**Supportive treatment:** / **Antibiotic:** ceftriaxone, azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/
		no	/		/	/	resolved	/

AlKhateeb 2020[15]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	1 month
Tham 2020[16]	4	no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details) **Antiviral drug:** / **Other treatment:** /	/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
		no	/		/	/	/	/
Cinar 2020[17]	1	yes	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), meropenem **Antiviral drug:** favipiravir **Other treatment:**	/	myelodysplastic syndrome (MDS/RAEB1 FAB subtype) during the treatment	/	/

				convalescent plasma product, tocilizumab					
Sarinoglu 2020[18]	2	yes	hemodialysis	**Supportive treatment:** / **Antibiotic:** ATT (no details), piperacillin/tazobactam **Antiviral drug:** favipiravir, hydroxychloroquine **Other treatment:** /	hemodialysis	/	/	/	
		no	/	**Supportive treatment:** / **Antibiotic:** ATT (no details), azithromycin, plaquanil, ciprofloxacine, tazocilli **Antiviral drug:** / **Other treatment:**	/	/	/	/	

				/				
Lopinto 2020[19]	1	yes	/	/	/	/	resolved	14
Musso 2021[20]	1	no	/	**Supportive treatment:** chest tube drainage **Antibiotic:** amikacin, moxifloxacin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hidroxicloroquine **Other treatment:** corticosteroids	/	/	died	14

Luciani 2020[21]	1	no	/	**Supportive treatment:** / **Antibiotic:** amoxicillin/clavulanic acid, clarithromycin, piperacillin/tazobactam, linezolid **Antiviral drug:** hydroxychloroquine, lopinavir, ritonavir **Other treatment:** paracetamol	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	> 12
Bouaré 2020[22]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), azithromycin **Antiviral drug:** chloroquine **Other treatment:** /	/	/	resolved	10

Essajee 2020[23]	1	no	/	**Supportive treatment:** high flow nasal oxygen **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** / **Other treatment:** prednisone, aspirin, dexamethasone	/	/	on treatment	/
Pillay 2021[24]	1	no	/	**Supportive treatment:** rebreather mask **Antibiotic:** ATT, amoxicillin/clavulanic acid, azithromycin, trimethoprim, sulfamethoxazole **Antiviral drug:** / **Other treatment:** pyridoxine, prednisone, prophylactic clexane, dexamethasone, vitamin D, zinc, thiamine, ascorbic acid, therapeutic anticoagulation	/	/	on treatment	/

Agada 2021[25]	1	no	/	**Supportive treatment:** supplemental oxygen **Antibiotic:** ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** no **Other treatment:** dexamethasone, acetaminophen, pyridoxine phosphate, zinc-sulphate, Vitamin C	ATT	/	resolved	14
Gbenga 2020[26]	2	no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir, ritonavir **Other treatment:** Vitamin C, Zinc sulfate, oral prednisolone, tabs pyridoxine	/	/	resolved	17

		no	/	**Supportive treatment:** oxygen therapy **Antibiotic:** azithromycin, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** lopinavir/ritonavir, **Other treatment:** vitamin C, zinc sulfate, tabs pyridoxine, IV hydrocortisone, clexane	/	/	died	6
Rivas 2020[27]	2	yes	/	**Supportive treatment:** invasive mechanical ventilatory support **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), levofloxacin, azithromycin, **Antiviral drug:** hydroxychloroquine, tenofovir, lamivudine, and dolutegravir **Other treatment:** low molecular weight heparin	/	nosocomial pneumonia	resolved	2 months

		no	rifampicin and isoniazid	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ceftriaxone, azithromycin, **Antiviral drug:** hydroxychloroquine **Other treatment:** enoxapari	/	/	resolved	7
Farias 2020[28]	2	no	/	**Supportive treatment:** oxygen via nasal cannula **Antibiotic:** ATT (isoniazid, rifampin, ethambutol, and pyrazinamide), ceftriaxone, azithromycin **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	21

		no	/	**Supportive treatment:** / **Antibiotic:** azithromycin, ceftriaxone, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	7
Pinheiro 2020[29]	1	no	/	/	/	/	/	/
Gerstein 2021[30]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (levofloxacin, rifampicin, isoniazid, ethambutol) **Antiviral drug:** hydroxychloroquine, **Other treatment:** convalescent plasma	ATT (rifampicin, isoniazid)	/	resolved	42

Cutler 2020[31]	1	no	/	**Supportive treatment:** nasal canula oxygen supplementation **Antibiotic:** ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide), **Antiviral drug:** hydroxychloroquine **Other treatment:** /	/	/	resolved	/
Sasson 2020[32]	1	yes	intubated for hypoxia	**Supportive treatment:** intubated for hypoxia **Antibiotic:** azithromycin, ceftriaxone, cefepime, vancomycin, piperacillin/tazobactam, meropenem, caspofungin, metronidazole, voriconazole, ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide) **Antiviral drug:** hydroxychloroquine **Other treatment:** dexamethasone, methylprednisolone, vitamin B6, metformin, alogliptin, long-acting insulin and rapid-acting insulin	/	/	resolved	> 35

Freij 2020[33]	1	no	amoxicillin	**Supportive treatment:** / **Antibiotic:** azithromycin **Antiviral drug:** hydroxychloroquine, remdesivir **Other treatment:** dexamethasone, antihypertensive medications	/	/	died	26
		no	/	/	/	/	resolved	
		no	/	/	/	/	resolved	
Vanzetti 2020[34]	3	no	/	**Supportive treatment:** oxygen support **Antibiotic:** / **Antiviral drug:** / **Other treatment:** corticosteroids	ATT (rifampicin, isoniazid, ethambutol, and pyrazinamide)	/	resolved	longer hospital stay than average

Vilbrun 2020[35]	1	no	/	**Supportive treatment:** / **Antibiotic:** ATT (bedaquiline, levofloxacin, linezolid, clofazimine, pyrazinimide) **Antiviral drug:** / **Other treatment:** /	/	/	resolved	3 weeks from the diagnosis

/: No information provided

Table 6 Detailed treatment information of included retrospective cohort studies (n=7)

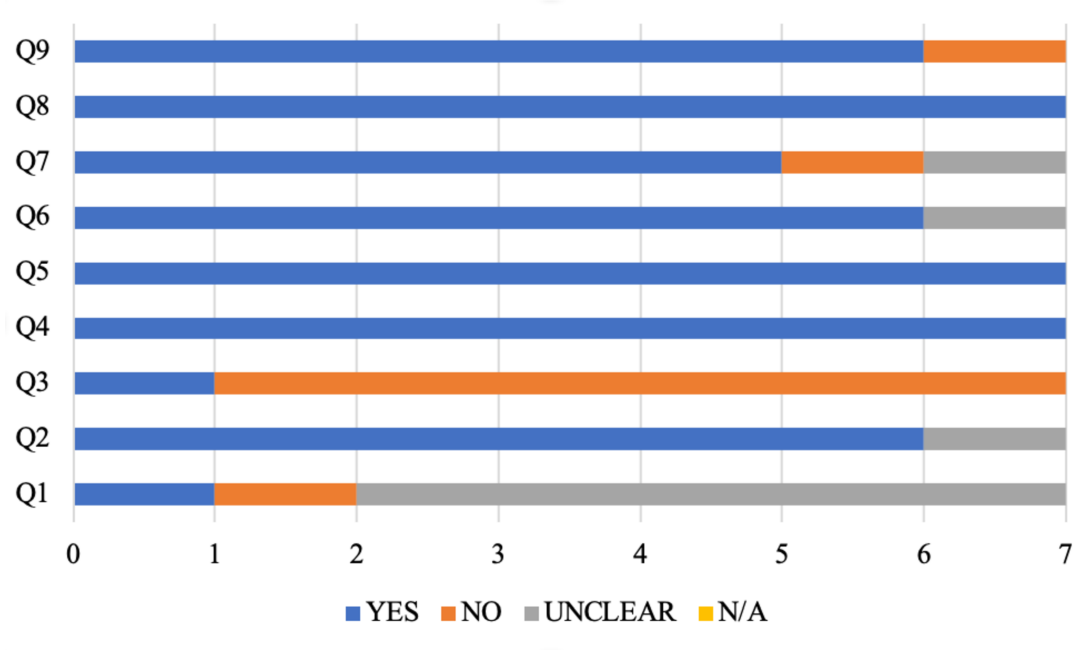
First author (year)	Country	Time	single center	Sample size	In-hospital	age	gender	Comorbidity	Died	Fatality rate (in-hospital or total)
Gupta 2020[36]	India	1 February 2020 to 14 June 2020	yes	22	yes	19-67	20 male (90.9%) 2 female (9.1%)	Hypertension 4 (18.2%) Diabetes 3 (13.6%) Seizure disorder 2 (9.1) Hypothyroidism 1 (4.5%)	6	27.3 (in-hospital)
Motta 2020[37]	Belgium, Brazil, France, Italy, Russia, Singapore,	12 March, 2020 to 5 May, 2020	no	49	no	27-70	/	/	7	14.3% (total)

	Spain, and Switzerland									
Sy 2020[38]	Philippines	May 17, 2020 to June 15, 2020	no	106	no	/	73 male (68.9%) 33 female (31.1%)	Hypertension 22 (20.8) Diabetes 14 (13.2) Cancer 1 (0.9) Renal cancer 4 (3.8) Cardiac disease 8 (7.5) Asthma 4 (3.8) COPD 3 (2.8)	25	23.6% (total)
				66	yes	/		Hypertension 16 (24.2) Diabetes 10 (15.2) Cancer 1 (1.5) Renal cancer 4 (6.1) Cardiac disease 7 (10.6) Asthma 1 (1.5) COPD 1 (1.5)	18	27.3% (in hospital)
Stochino 2020[39]	Italy	/	yes	20	yes	27-47	12 male (60%) 8 female (40%)	/	1	5% (in-hospital)
Davies 2020[40]	South Africa	Till to 9 June, 2020	no	2128	no	20 or above	/	/	113	5.3% (total)
			no	469	yes	20 or	/	/	102	21.7%

						above				(in-hospital)
Domingo 2020[41]	Argentina	1 March to 30 June, 2020	yes	23	yes	5-82	18 male (78.3%) 5 female (21.7%)	Smoking and other addictions 16(69.6) HIV 4(17.4) Psychosis 1(4.3) Pulmonary thromboembolism 1(4.3) Arterial hypertension 1(4.3)	2	8.7% (in-hospital)
Gubkina 2020[42]	Russia	March to June, 2020		24	yes	/	/	/	0	0% (in-hospital)

/: No information provided

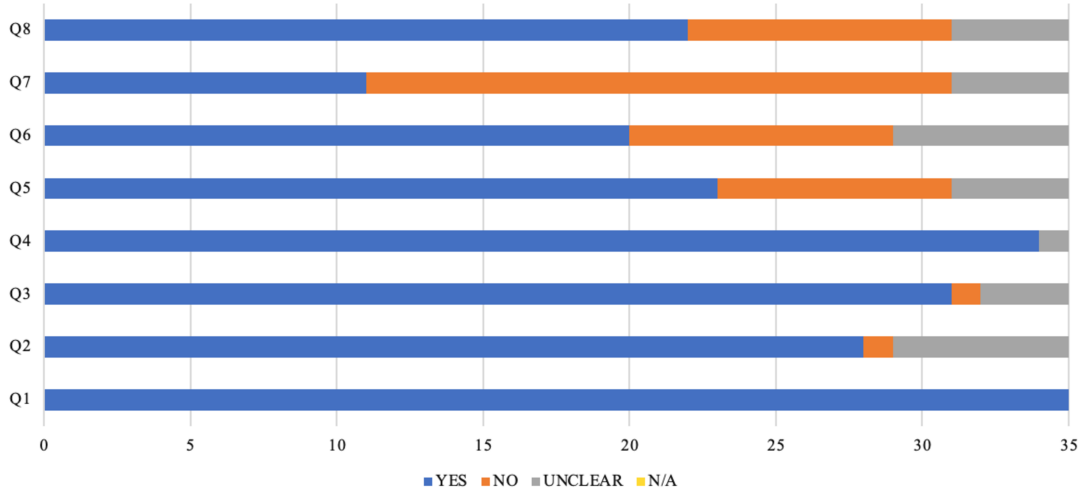
Figure 1 The results of quality assessment for prevalence study



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the

identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

Figure 2 The results of quality assessment for case report



Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly

described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 7 Quality assessment of included case reports (n=35)

Study (Year) \ Results	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Agada 2021	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Yao 2020	Yes	Yes	Yes	Unclear	Unclear	No	No	No
Yadav 2020	Yes	Yes	Yes	Yes	No	No	No	No
Wang 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Vilbrun 2020	Yes	Yes	Yes	Yes	No	Unclear	No	Yes
Tham 2020	Yes	Yes	Yes	Yes	No	No	No	No
Singh 2020	Yes	Yes	Yes	Yes	Unclear	No	Yes	Yes
Sasson 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear
Sarma 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Sarinoglu 2020	Yes	Yes	Yes	Yes	Unclear	Yes	No	Yes
Rivas 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Pinheiro 2020	Yes	Yes	Unclear	Yes	No	No	No	No
Pillay 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Musso 2021	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Luciani 2020	Yes	Yes	Yes	Yes	Yes	No	No	Yes

Lopinto 2020	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes
Liu 2020	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Kumar 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Khayat 2021	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Gbenga 2020	Yes	Yes	Yes	Yes	Unclear	Unclear	Unclear	No
Freij 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cutler 2020	Yes	Yes	Unclear	Yes	Yes	Yes	No	Yes
Cinar 2020	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Cao 2020	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farias 2020	Yes	Yes	Yes	Yes	Yes	Unclear	No	Yes
Bouare 2020	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Baskara 2021	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
AlKhateeb 2020	Yes	Yes	Yes	Yes	No	Yes	No	Unclear

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Were patient's demographic characteristics clearly described? Q2: Was the patient's history clearly described and presented as a timeline? Q3: Was the current clinical condition of the patient on presentation clearly described? Q4: Were diagnostic tests or assessment methods and the results clearly described? Q5: Was the intervention(s) or treatment procedure(s) clearly described? Q6: Was the post-intervention clinical condition clearly described? Q7: Were adverse events (harms) or unanticipated events identified and described? Q8: Does the case report provide takeaway lessons?

Table 8 Quality assessment of included retrospective cohort studies (n=7)

Results / Study (Years)	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
Gupta 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Motta 2020	Unclear	Unclear	No	Yes	Yes	Yes	Yes	Yes	Yes

Sy 2020	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes	No
Stochino 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Davies 2020	Unclear	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Domingo 2020	Unclear	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Gubkina 2020	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes

Note: Four options for each question: yes, no, unclear and not appliable. Q1: Was the sample frame appropriate to address the target population? Q2: Were study participants sampled in an appropriate way? Q3: Was the sample size adequate? Q4: Were the study subjects and the setting described in detail? Q5: Was the data analysis conducted with sufficient coverage of the identified sample? Q6: Were valid methods used for the identification of the condition? Q7: Was the condition measured in a standard, reliable way for all participants? Q8: Was there appropriate statistical analysis? Q9: Was the response rate adequate, and if not, was the low response rate managed appropriately?

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